

# PAPER\_697: NGC 2525 with SN 2018gv: Type Ia Supernova Light Curve and UQFF

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## Abstract

NGC 2525 is a barred spiral galaxy (SBc) at 60 Mly hosting Type Ia SN 2018gv, observed by Hubble in time-lapse. The Arnett light curve model and Phillips relation are applied with UQFF modifications to the peak luminosity.

## 1. Parameters

| Parameter     | Value                        |
|---------------|------------------------------|
| $M_{galaxy}$  | $3 \times 10^{10} M_{\odot}$ |
| Distance      | 60 Mly                       |
| $L_{SN,peak}$ | $1.5 \times 10^{43}$ W       |
| $M_{Ni56}$    | $0.6 M_{\odot}$              |
| Rise time     | 14 days                      |

## 2. Type Ia Light Curve (Arnett)

$$L(t) \approx L_{peak} e^{-\frac{1}{2} \left( \frac{t-t_r}{\sigma_t} \right)^2} \times e^{-(t-t_r)/t_d}$$

## 3. Phillips Relation

$$M_B \approx -19.3 + 0.74(\Delta m_{15} - 1.1)$$

$\Delta m_{15}$ : magnitude decline in first 15 days post-peak.

## 4. UQFF Luminosity Correction

$$L_{UQFF}(t) = L_{SN}(t) \left( 1 + \frac{\rho_{SCm}}{\rho_{UA}} \right) (1 - f_{TRZ})$$

## 5. Barred Disk Rotation

$$v_c(R) = \sqrt{v_{disk}^2 + v_{bar}^2}$$

where  $v_{bar}^2 = GM_{gal} \cdot 0.3(1 - e^{-R/R_{bar}})/R$ .

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                        | SM / Experiment                        | Source                              | Alignment    |
|----------------------------------|------------------------------------------------------------------------|----------------------------------------|-------------------------------------|--------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                       | 1/137.036                              | PDG 2024                            | ✓ Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)                | $1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018                         | ✓ Consistent |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$     | Super-K 2024                        | ✓ Consistent |
| UQFF buoyancy signature          | F_U_Bi_i unique gravitational correction                               | Not yet measured                       | Future gravitational wave detectors | Testable     |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F\_U\_Bi\_i) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

### References

- Arnett (1982), ApJ, 253, 785
- Phillips (1993), ApJ, 413, L105
- UQFF Framework, Star-Magic Session 174

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\*Whitepaper auto-generated by \_gen\_whitepapers\_688\_701.py — Star-Magic Session 174\*